

Software Requirements Specification (SRS)

AI Resume Analyzer

1. Introduction

1.1 Purpose

The AI Resume Analyzer is a web-based application that assists job seekers by analyzing uploaded resumes using Artificial Intelligence and recommending suitable job opportunities. The system leverages AI Agents and Retrieval-Augmented Generation (RAG) to evaluate resumes, identify strengths and weaknesses, and provide personalized career recommendations.

1.2 Project Objective

- Analyze resumes automatically.
- Recommend skills required for target jobs.
- Provide AI-generated career advice.
- Demonstrate the integration of AI Agents with a modern web application.

2. Technology Constraints

Students must implement using only:

Backend: Python, FastAPI

Database: SQLite

Frontend: HTML, CSS, Vanilla JavaScript

Note: No React, Angular, Vue, or other frameworks.

3. Functional Requirements

FR-1 User Authentication: Register, Login, Logout

FR-2 Resume Upload: Upload PDF or DOCX resumes.

FR-3 Resume Analysis: Extract information, detect technical/soft skills, identify education & experience, generate summary.

FR-4 Job Recommendation: Compare resumes with jobs, calculate matching score, recommend jobs, explain recommendations.

FR-5 Resume Improvement: Missing skills, weaknesses, improvements, certifications, learning resources.

FR-6 Job Management: Add, Edit, Delete, View jobs.

FR-7 Job Search: Search jobs using different criteria.

4. AI Requirements

Resume Analyzer Agent: Analyze resumes, extract information, generate summaries.

Job Matching Agent: Compare resumes with job descriptions, calculate similarity scores, recommend jobs.

Career Advisor Agent: Recommend courses, certifications, missing skills, answer career questions.

5. RAG Requirements

Retrieve relevant information from a knowledge base before generating responses.

Knowledge Base: Job descriptions, Skill descriptions, Career roadmaps, Learning resources, Resume writing guidelines.

6. Non-Functional Requirements

User-friendly interface.

Validate user inputs.

Handle invalid files gracefully.

Reasonable response time.

REST API principles.

Meaningful error messages.

Readable and modular code.

7. Deliverables

Complete source code.

API documentation.

Demo.

ER Diagram.

8. Evaluation Criteria (100 Marks)

Criterion	Marks
Backend Development (python + FastAPI)	20
Frontend Development (HTML, CSS, JavaScript)	10
Database Design (SQLite)	10
Resume Analysis	15
AI Agents	20
RAG Implementation	10
Code Quality & Error Handling	10
Documentation	5
Total	100